Natural Language Processing Assignment 2

Report

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Sequence Modeling, Seq2Seq Translation, and Attention-based NMT

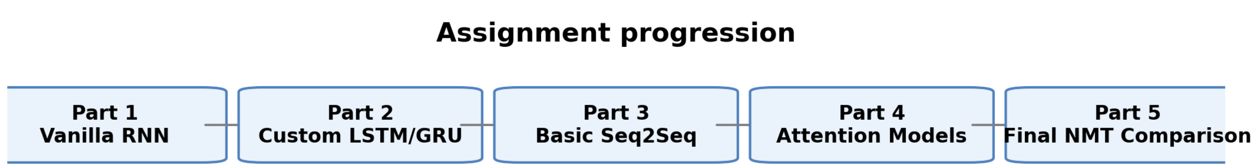


Figure 1. High-level progression of the assignment from basic language modeling to final NMT comparison.

2. Part 1 - Sequence Modeling with Vanilla RNN

2.1 Dataset preprocessing (P1T1)

The WikiText-2-v1 corpus was normalized, tokenized, converted into a vocabulary, and prepared for next-word language modeling. The key dataset statistics are shown below.

Metric	Value
Vocabulary Size	25,000
Total Tokens	2,007,146
Average Sentence Length	114.33
Training Sentences	17,556
Validation Sentences	1,841
Test Sentences	2,183

2.2 Vanilla RNN training and evaluation (P1T2)

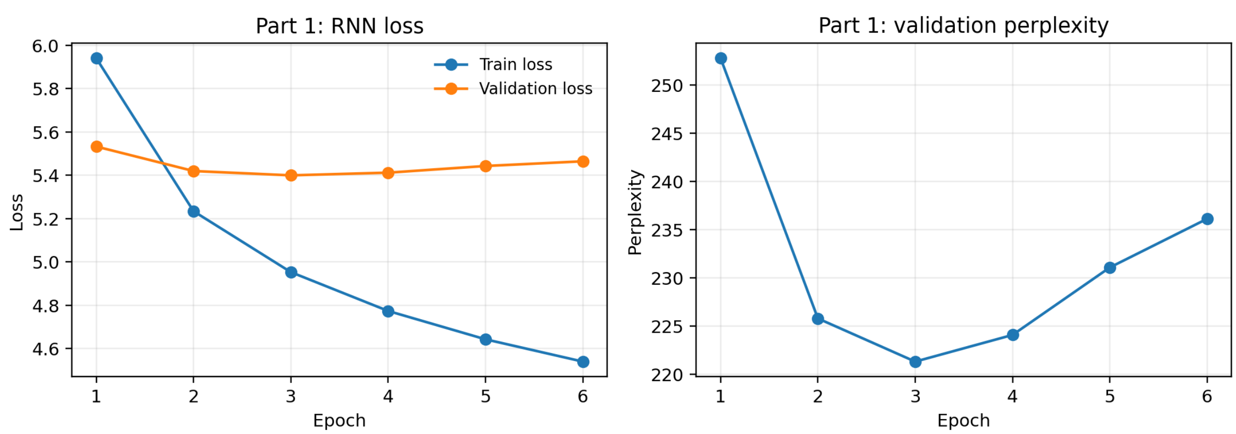


Figure 2. Training loss, validation loss, and validation perplexity for the vanilla RNN model.

Metric	Value
Final Validation Loss	5.46
Final Validation Perplexity	236.12
Test Loss	5.40
Test Perplexity	221.53
Training Time (sec)	441.74

Training loss decreased steadily, which shows that the model learned useful local sequence patterns. Validation perplexity improved early and then began to rise again, which indicates that the model reached its practical limit and did not generalize as strongly as the later gated models.

2.3 Sequence generation (P1T3)

Twenty sequences were generated from the trained model. The report shows only a few representative examples to keep the document concise; the full set remains in the notebook.

Generated example 1

Source: natural language

Prediction: natural language . The city operates by <unk> ...

Generated example 4

Source: artificial intelligence

Prediction: artificial intelligence , and <unk> . The main...

Generated example 8

Source: the history of

Prediction: the history of a highway , the <unk> Bridge , ...

Perplexity was used as the main evaluation metric because it directly measures how well the model predicts the next token. The generated samples were also inspected qualitatively for fluency, repetition, coherence, and the frequency of <UNK> tokens.

3. Part 2 - Manual LSTM and GRU Sequence Models

3.1 Dataset preparation and training setup (P2T1-P2T4)

Both the LSTM and GRU models were implemented manually using basic tensor operations and trained under the same preprocessing pipeline and sequence length. This ensured a fair architecture comparison.

Metric	Value
Vocabulary Size	25000
Train Samples	210329
Validation Samples	21931
Test Samples	24821
Sequence Length	20

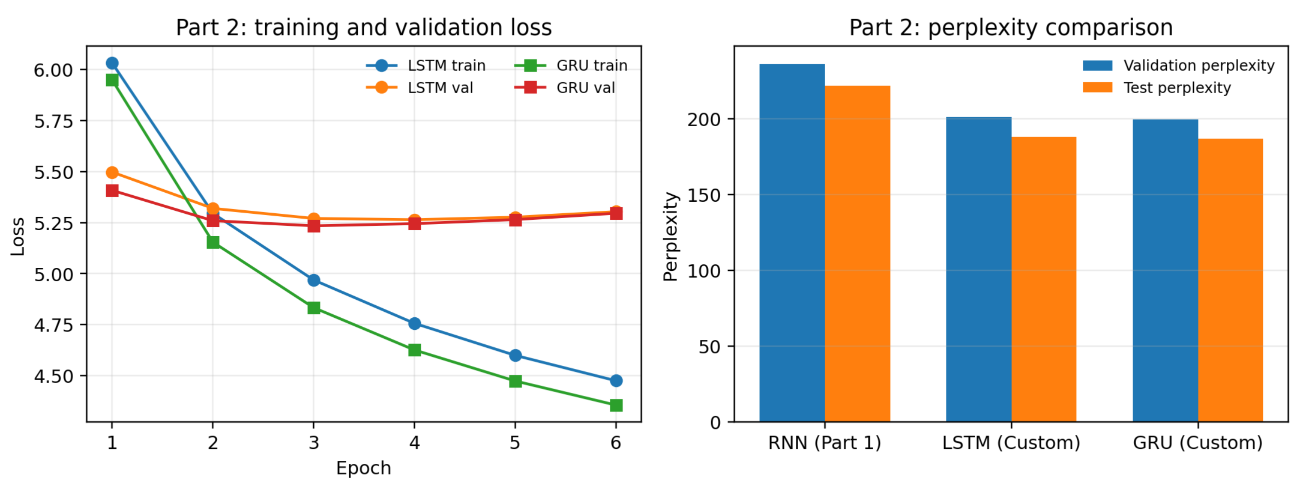


Figure 3. Loss behavior and perplexity comparison for the custom LSTM and GRU models, alongside the Part 1 RNN baseline.

Both gated models converged better than the vanilla RNN. GRU reached the best overall validation and test perplexity, while also training slightly faster than the custom LSTM.

Model	Validation Perplexity	Test Perplexity	Training Time (sec)
RNN (Part 1)	236.12	221.53	441.74
LSTM (Custom)	201.12	188.04	776.89
GRU (Custom)	199.57	186.97	758.39

3.2 Next-word prediction and gate analysis (P2T5-P2T7)

The next-word prediction outputs showed that both gated models handled short contexts better than the vanilla RNN. Gate activations were also examined to understand how information was controlled internally.

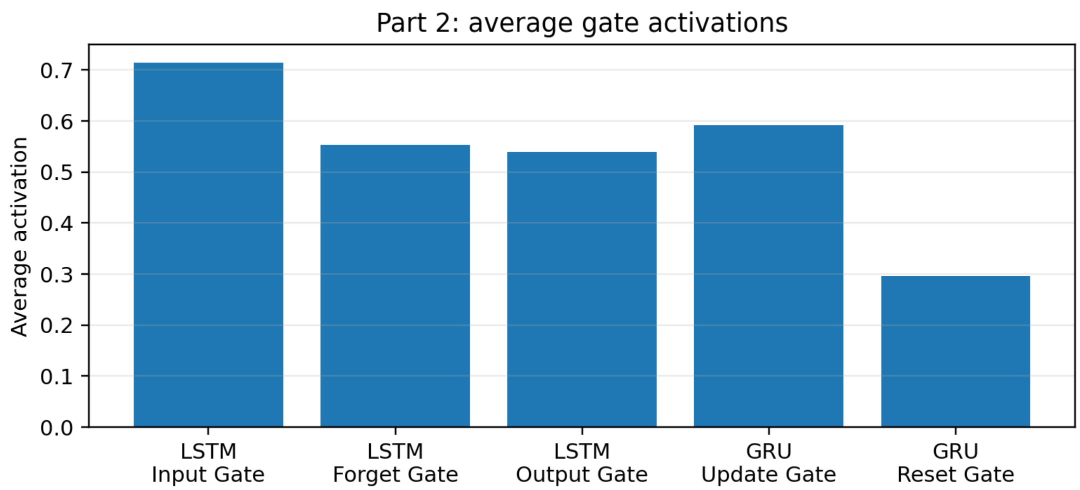


Figure 4. Average gate activations recorded from a selected validation sentence.

Model	Gate	Average Activation
LSTM	Input Gate	0.7140
LSTM	Forget Gate	0.5530
LSTM	Output Gate	0.5392
GRU	Update Gate	0.5914
GRU	Reset Gate	0.2954

The LSTM input gate stayed relatively open, which means the model frequently incorporated new information. In GRU, the update gate was more active than the reset gate, which suggests the model often preferred to preserve useful context rather than discard it.

3.3 Long-context prediction and final comparison (P2T8)

Model	Predicted Word	Confidence
RNN	.	0.3175
LSTM	.	0.2606
GRU	.	0.6654

In the long-context prediction test, GRU produced the strongest confidence score. Overall, the final ranking for Part 2 was GRU first, LSTM second, and the vanilla RNN third.

4. Part 3 - Basic Seq2Seq Encoder-Decoder Translation

4.1 Model pipeline and experiment summary (P3T1)

A basic encoder-decoder model was implemented for English-to-German translation on the Multi30k dataset. The figure below shows the translation flow used in the model.

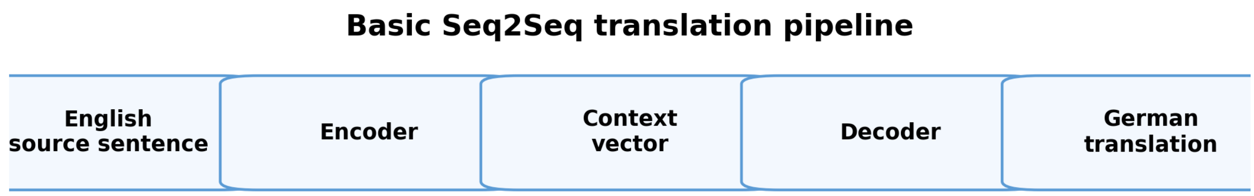


Figure 5. Basic Seq2Seq translation pipeline used for Part 3.

Item	Value
Train pairs	29,000
Validation pairs	1,000
Test pairs	1,000
Source vocabulary	7,704
Target vocabulary	9,597
Final train loss	2.33
Final validation loss	3.93
Training time (min)	4.12
BLEU score	21.34
Token accuracy	0.31

4.2 Translation generation and evaluation (P3T2-P3T3)

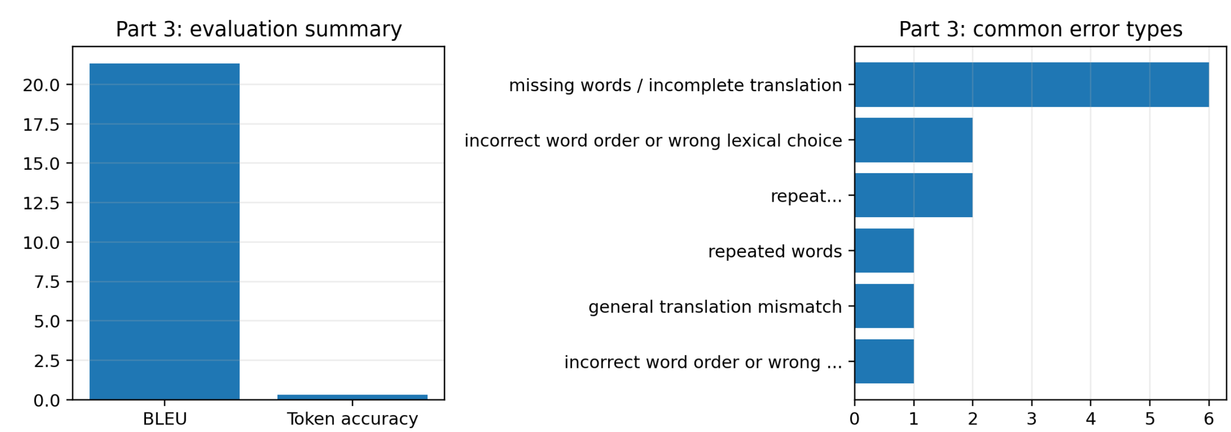


Figure 6. Basic evaluation summary and common translation error types for the Part 3 model.

The basic Seq2Seq model achieved moderate translation quality. BLEU reached 21.34 and token-level accuracy reached 0.3131, which indicates that the model learned useful source-target mappings but still struggled with lexical precision and completeness.

Selected Part 3 translation example 1

Source: a man in an orange hat starring at something.

Reference: ein mann mit einem orangefarbenen hut, der etw...

Prediction: ein mann in einem orangefarbenen sicherheitswe...

Selected Part 3 translation example 9

Source: a guy works on a building.

Reference: ein typ arbeitet an einem gebäude.

Prediction: ein mann arbeitet an einem <UNK>

4.3 Error analysis (P3T4)

Error type	Count
missing words / incomplete translation	6
incorrect word order or wrong lexical choice	2
repeat...	2
repeated words	1
general translation mismatch	1
incorrect word order or wrong ...	1

The most frequent weak outputs involved missing words, incomplete translations, wrong lexical choices, and word-order issues. This is expected because a basic Seq2Seq model compresses the source sentence into a single context vector, which becomes a bottleneck on longer or more complex examples.

5. Part 4 - Attention Mechanism in Translation

Dataset note: For this part, a reduced WMT14 subset was used: 100,000 training pairs, 1,500 validation pairs, and 1,000 test pairs. This should be interpreted as a resource-constrained experimental approximation of the assignment's 500K–1M target rather than a full-scale implementation.

5.1 Attention architecture and mechanisms (P4T1)

Three attention mechanisms were implemented: Bahdanau attention, Luong attention, and scaled dot-product attention. In all cases, attention scores are computed between decoder states and encoder states, normalized with softmax, and used to build a context vector that is combined with the decoder.

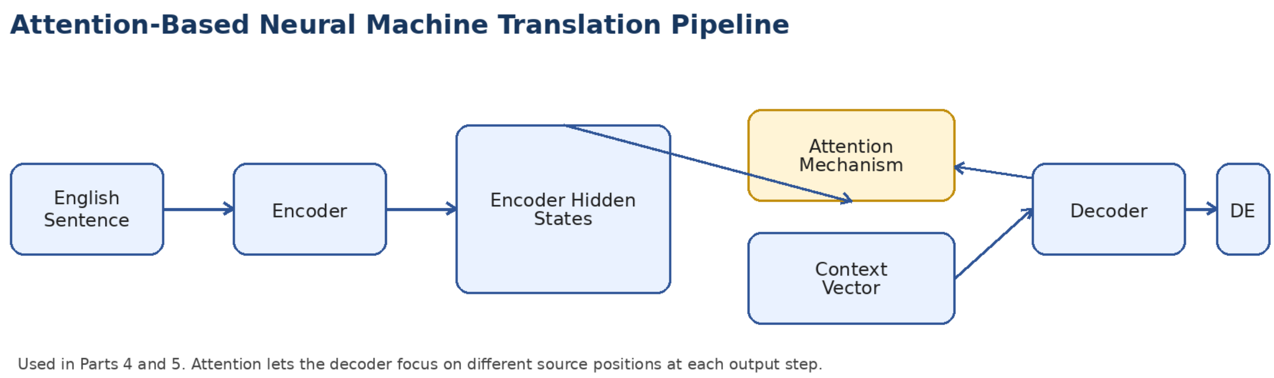
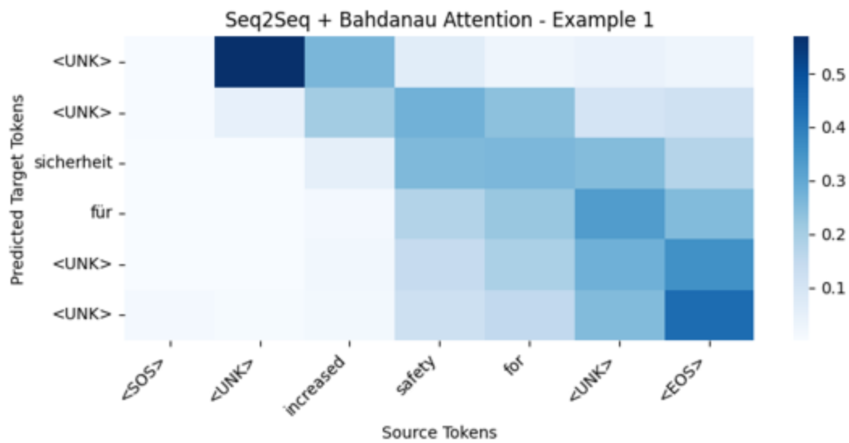


Figure 7. Attention-based neural machine translation pipeline used in Parts 4 and 5.

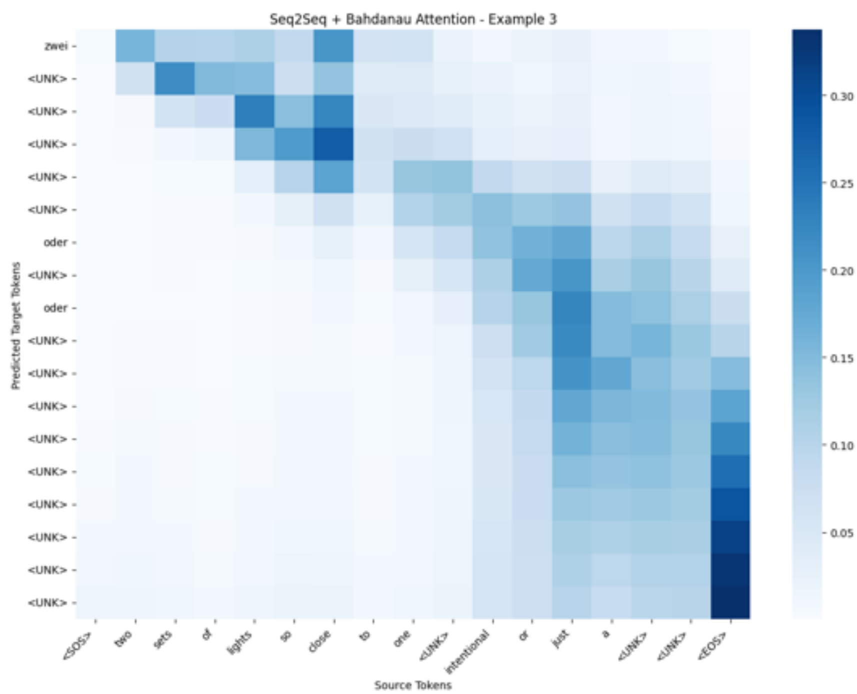
5.2 Attention visualization (P4T2)

Attention heatmaps are important because they show which source tokens influenced each predicted target token. A smaller sample is shown here in the body of the report, while the full collage of 15 examples is placed in the appendix.

Example 1



Example 3



Example 10

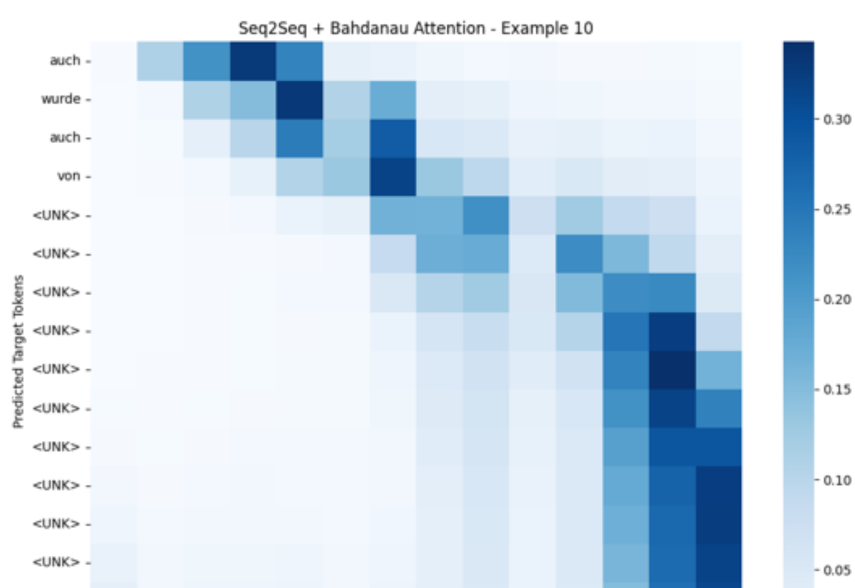


Figure 8. Sample attention heatmaps from the Bahdanau model.

5.3 Attention model comparison (P4T3)

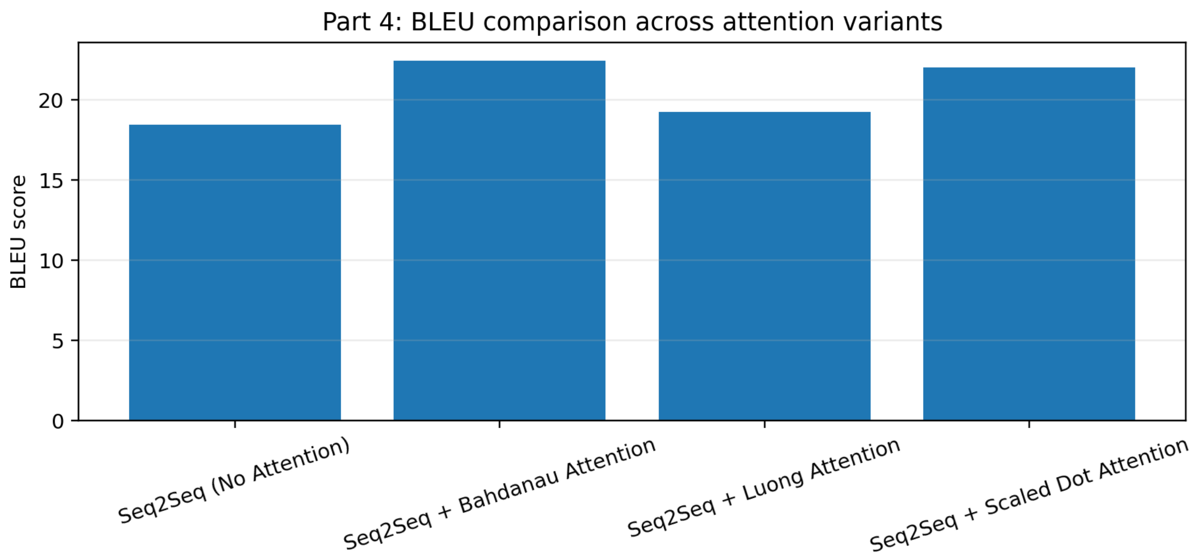


Figure 9. BLEU score comparison for no-attention and attention-based models.

Model	BLEU Score	Final Validation Perplexity	Training Time (min)
Seq2Seq (No Attention)	18.45	264.18	10.06
Seq2Seq + Bahdanau Attention	22.45	193.42	15.47
Seq2Seq + Luong Attention	19.26	207.79	14.28
Seq2Seq + Scaled Dot Attention	22.02	222.12	14.05

Bahdanau attention performed best in this notebook, achieving the highest BLEU score and the lowest validation perplexity. Overall, attention improved translation quality by allowing the decoder to focus on the most relevant source positions during generation.

6. Part 5 - Final Neural Machine Translation System

6.1 Final setup and experimental configuration (P5T1)

For the final Part 5 experiment, the notebook used a larger 300K WMT14 configuration and restored the best validation-loss checkpoint for each model before evaluation. The pipeline also preserved raw source and reference text for qualitative examples and kept checkpoint-based recovery enabled for long training runs.

Item	Value
Train pairs kept	300,000
Train pairs skipped during preprocessing	98,389
Validation pairs kept	2,637
Validation pairs skipped	363
Test pairs kept	2,000
Test pairs skipped	282
Source vocabulary	30,000
Target vocabulary	30,000
Max sequence length	40
Epochs	3
Batch size	32
Embedding dimension	256
Hidden dimension	512
Dropout	0.30
Learning rate	0.0008
Teacher forcing	0.70 → 0.40
Checkpointing	Enabled (resume + save every epoch)
BERTScore evaluation limit	500 samples

Interpretation note: Part 4 uses a 100K WMT14 configuration, whereas Part 5 uses the larger 300K configuration summarized below. For that reason, cross-part numeric comparisons should be interpreted cautiously; the Part 5 tables represent the final NMT experiment under the stronger resource-constrained setup.

6.2 Final model comparison (P5T1-P5T2)

All five required final models were trained: Seq2Seq RNN, Seq2Seq LSTM, Seq2Seq GRU, Seq2Seq LSTM + Bahdanau Attention, and Seq2Seq GRU + Bahdanau Attention. The final comparison table is shown below.

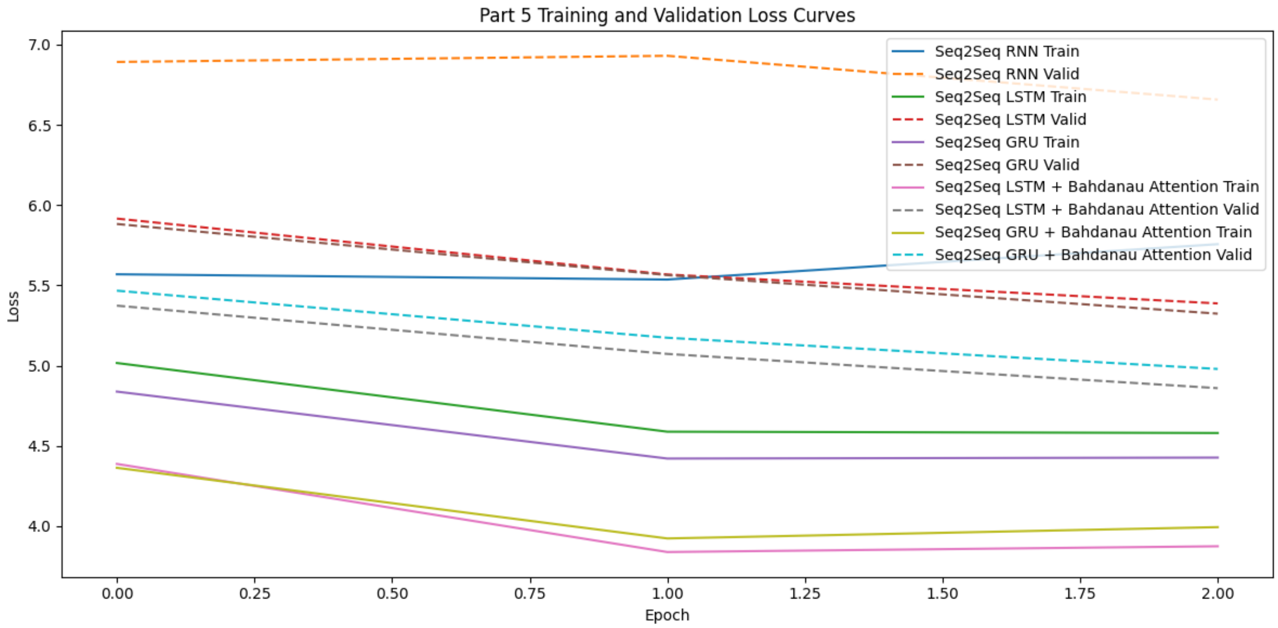
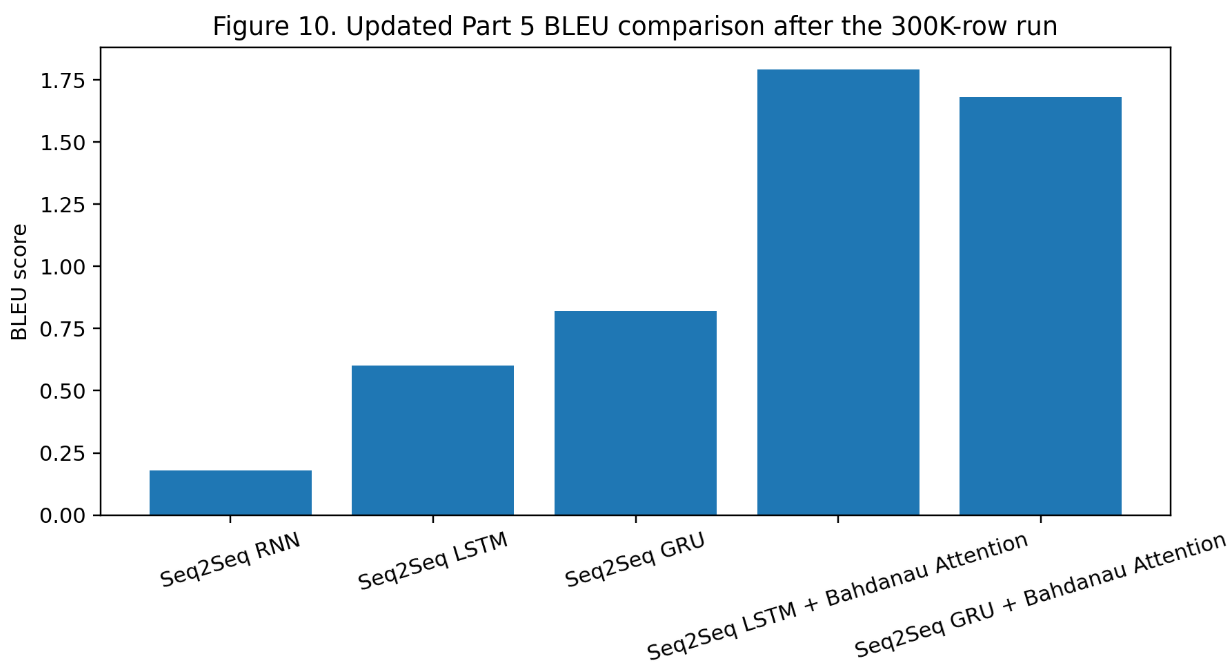


Figure 10. Part 5 training and validation loss curves from the final 300K notebook run.



Model	BLEU	BERTScore F1	Sentence Accuracy (Exact Match)	METEOR	Training Time (min)
Seq2Seq RNN	0.18	0.5785	0.0	0.0891	76.65
Seq2Seq LSTM	0.6	0.5988	0.0	0.1144	81.36
Seq2Seq GRU	0.82	0.6075	0.0	0.1271	79.97
Seq2Seq LSTM + Bahdanau Attention	1.79	0.6438	0.0005	0.1769	131.79
Seq2Seq GRU + Bahdanau Attention	1.68	0.6453	0.0	0.1723	129.24

The plain Seq2Seq RNN remained the weakest model by a large margin. Both non-attention gated models improved over the RNN baseline, and both Bahdanau-attention models produced the strongest translation quality overall. In the final 300K configuration, Seq2Seq LSTM + Bahdanau Attention achieved the highest BLEU (1.79), METEOR (0.1769), and sentence accuracy (0.0005), while Seq2Seq GRU + Bahdanau Attention achieved the best BERTScore F1 (0.6453).

Note: BERTScore F1 was computed on a capped subset of 500 test predictions for runtime control. Sentence accuracy denotes exact-match sentence proportion on the evaluated test set.

The expanded training run greatly increased computation time. The attention models required around 129-132 minutes each, compared with roughly 77-81 minutes for the non-attention baselines.

6.3 Optimization summary and interpretation

Model	Best Valid Loss	Final Valid Perplexity	Final Train Acc	Final Valid Acc	Training Time (min)
Seq2Seq RNN	6.6576	778.65	0.1594	0.0876	76.65
Seq2Seq LSTM	5.388	218.76	0.2425	0.1618	81.36
Seq2Seq GRU	5.3242	205.24	0.2423	0.1569	79.97
Seq2Seq LSTM + Bahdanau Attention	4.8601	129.04	0.2923	0.1795	131.79
Seq2Seq GRU + Bahdanau Attention	4.9798	145.44	0.2728	0.1691	129.24

The low absolute BLEU values reflect a more difficult and larger final configuration, including a larger vocabulary, longer decoding length, bigger test set, and only three training epochs. Under these stricter conditions, the relative ranking is the most important result: attention still helps, and LSTM/GRU-based encoder-decoder models remain clearly stronger than the plain RNN.

6.4 Translation examples by roll-number batches (P5T3)

Because the roll number is 25L-8021, the report includes translation examples in three batches based on the digits 8, 2, and 1. This produces 11 total examples (8 + 2 + 1), reported in the appendix. The final Part 5 qualitative examples shown below use Seq2Seq LSTM + Bahdanau Attention, the strongest BLEU model in this report.

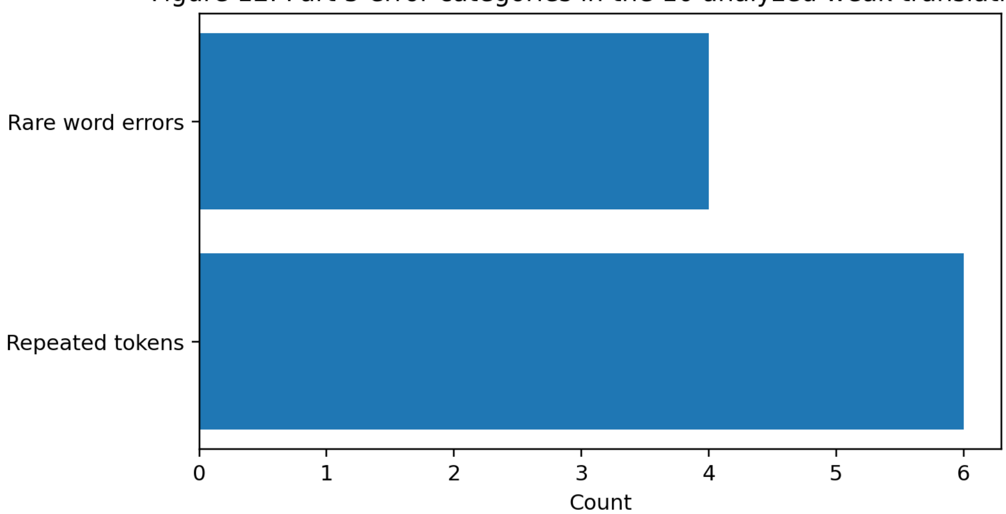
Batch Group	Source Sentence	Reference Translation	Model Prediction
Digit 8 batch	gutach: increased safety for pedestrians	gutach: noch mehr sicherheit für fußgänger	<UNK>: höhere sicherheit für fußgänger
Digit 2 batch	"at times of high road and pedestrian traffic,...	"bei dem hohen verkehrs- und fußgängeraufkomme...	" <UNK> und <UNK> <UNK> <UNK> <UNK> zusätzlich...
Digit 1 batch	therefore the construction of an additional se...	daher sei der bau einer weiteren ampel mehr al...	die <UNK> zusätzliche zusätzliche <UNK> war al...

These examples show a mixed picture. The model often captures part of the meaning or some important content words, but repetition and <UNK>-driven degradation still appears in difficult cases.

6.5 Error analysis (P5T4)

Ten weak translations from the best model were analyzed. In this 10-sample error set, repeated-token and rare-word failures dominated, while other categories were much less visible than in earlier smaller-scale runs.

Figure 12. Part 5 error categories in the 10 analyzed weak translations



Error Type	Count
Rare word errors	4
Repeated tokens	6

Repeated tokens appeared in 6 of the 10 analyzed weak examples, while rare-word errors appeared in the remaining 4. Attention still improved the outputs by giving the decoder better access to source positions, but it could not fully resolve vocabulary coverage problems or repetition loops under the current compute budget.

6.6 Important Note: Computational limits

The assignment brief asks for 500K–1M WMT14 sentence pairs in Parts 4 and 5. In practice, the executed notebook used 100K training pairs for Part 4 and a 300K training-pair configuration for Part 5. These experiments should therefore be read as resource-constrained approximations of the full target setup. The limitation is stated explicitly so the evaluation remains academically honest and reproducible. Additional practical constraints also mattered:

- Only 3 epochs were used in Part 5, so convergence is incomplete relative to the scale of the task.
- BERTScore F1 was computed on a capped subset of 500 predictions for runtime control.
- Checkpointing and resume support were enabled so that the best model state could be recovered reliably after long runs.
- The attention models were significantly slower than the non-attention baselines, which limited how far the final experiments could be scaled in Kaggle.

The executed notebook also exports final CSV files for model comparison, translation examples, and error analysis, as well as a loss-curve image for reporting.

7. Overall conclusion

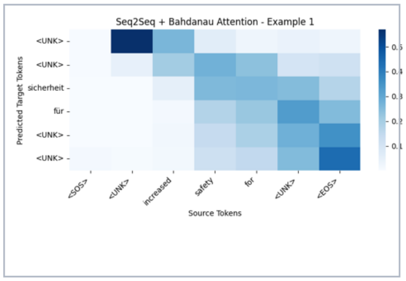
The assignment demonstrates a clear architectural progression: vanilla RNN -> gated recurrence -> basic Seq2Seq -> attention -> final NMT comparison. The core findings remain consistent across the report. Vanilla RNN is the weakest overall model. Manual GRU and LSTM significantly improve next-word prediction and long-context handling. Basic Seq2Seq translates simple patterns but struggles with completeness and word order. Attention improves translation quality by removing the fixed-context bottleneck. In the final 300K Part 5 configuration, both Bahdanau-attention variants remain the strongest overall systems.

Among the final Part 5 models, Seq2Seq LSTM + Bahdanau Attention is the best all-round model in this report because it gives the strongest BLEU, METEOR, and sentence accuracy, while Seq2Seq GRU + Bahdanau Attention remains highly competitive and gives the best BERTScore F1. The main limitation is still computational scale: even after upgrading Part 5 to 300K rows, the experiments remain below the assignment’s full 500K–1M target.

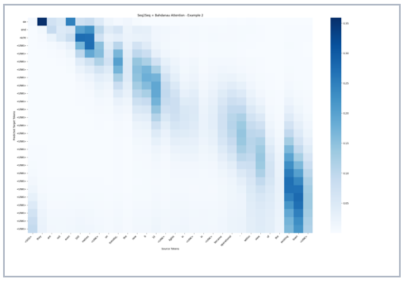
Appendix - Full Attention Heatmap Collage

The collage below includes the full set of 15 Bahdanau attention visualizations referenced in Part 4. It is placed in the appendix to keep the main body focused on the most important results.

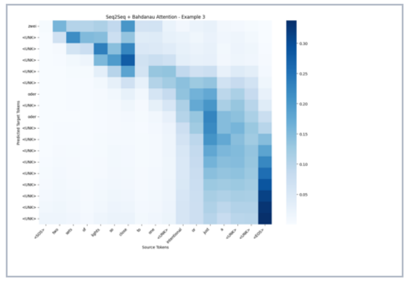
Appendix Heatmaps: Bahdanau Attention (15 Translation Examples)



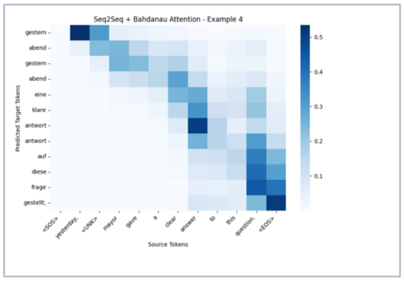
Example 1



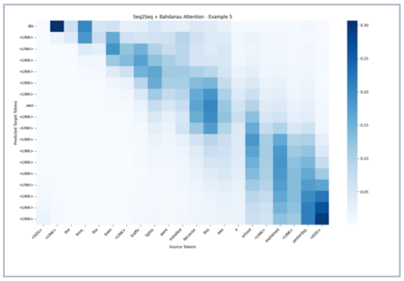
Example 2



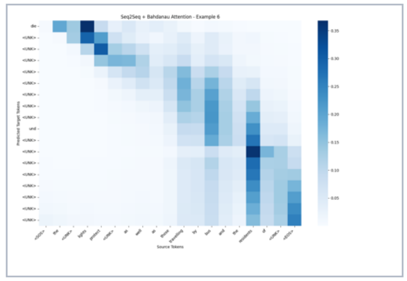
Example 3



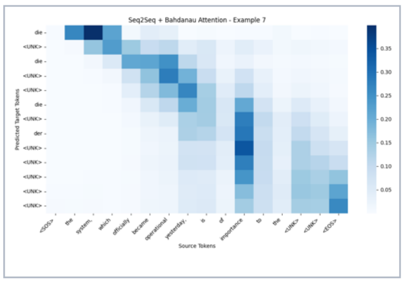
Example 4



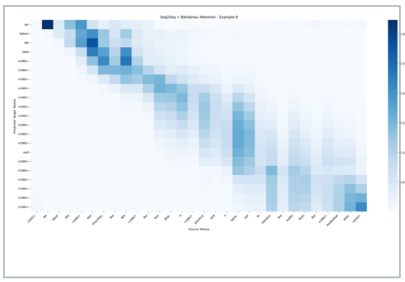
Example 5



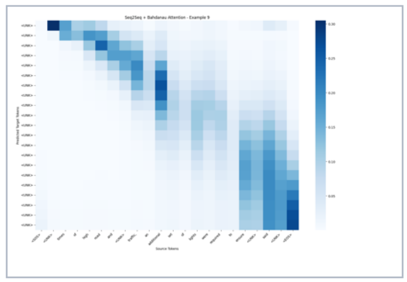
Example 6



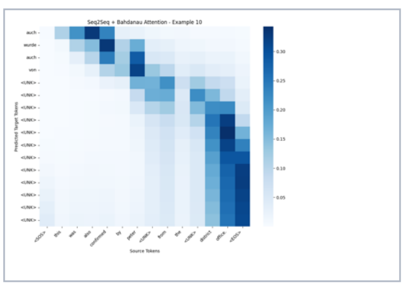
Example 7



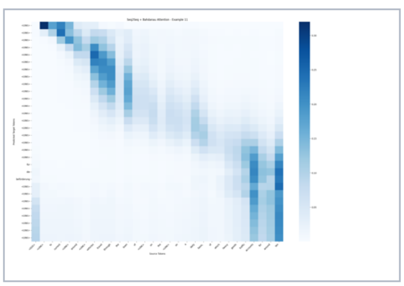
Example 8



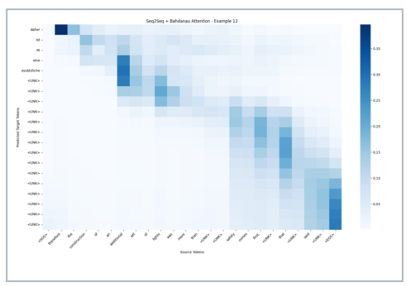
Example 9



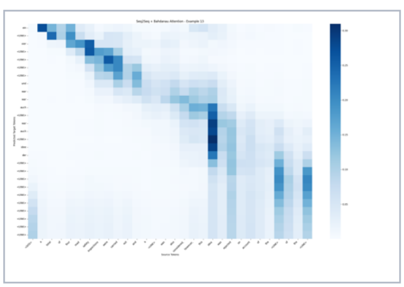
Example 10



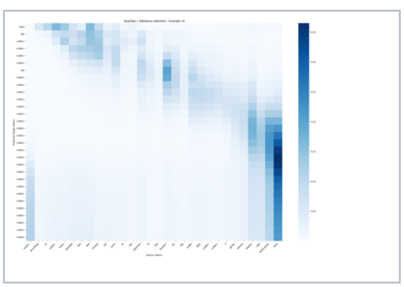
Example 11



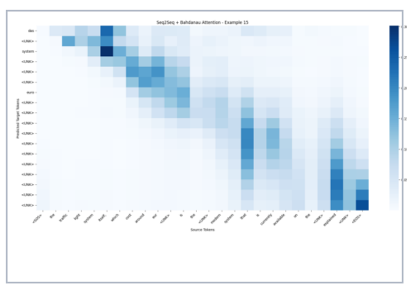
Example 12



Example 13



Example 14



Example 15

Appendix Figure A1. Full collage of 15 attention heatmaps.